AUFIT & BatchAufit
User's manual

By Jean Louis-Guérin (DrCoolZic)
Revision 1.0 - May 2015

Table of Content

AUFIT User's manual 1
Table of Content 2
Presentation 3
Program Download / Installation 3
Running the AUFIT Program 4
Using Aufit 5
- Loading Image Files 5
- Selecting a specific track 7
- Selecting a specific revolution 7
- The Histogram Graph 8
- The Information tab 8
- The Track View 9
  - Flux Transitions Chart 9
  - Bit Width Chart 9
  - Data Graph 10
  - Zooming and Panning in the Track view 10
- The Disk View 11
- The Data View 12
  - Track Layout 12
  - Track Buffer 13
  - Sectors Buffer 14
  - Saving Data View content 15
- The Analysis View 16
- Save Image View 17

Checking Quality of generated images 18
- Cleaning 20
- Analyzing Images 20

Document History 23
Program History 23

Presentation

This document describes the usage of the **AUFIT** program. AUFIT is the acronym for ***Atari Universal Floppy-Disk Image Tool***.

The program reads several Floppy Disk image's formats used with the Atari platform, including images generated at the flux level by devices like SuperCard Pro, KryoFlux. Once loaded the image is analyzed and its content is displayed in several Graphical and Text windows. It is then possible to save the loaded image using different output format.

Therefore the primary usage of Aufit program is to convert Stream files to a format suitable for Atari emulators. It is for example possible to convert SuperCard Pro SCP or KryoFlux RAW stream files to the Pasti STX format used by many Atari emulators. Beside the multi-format converter capability Aufit can be used to perform deep analysis of a disk image content. For example it is possible to analyze the key disk protections used or the quality of the input stream.

The program has been designed to be as simple as possible for casual users only interested in converting files. However the program also provides powerful features to deeply analyze floppy content for advanced users.

Program Download / Installation

The program can be downloaded from http://info-coach.fr/atari/software/pc-projects/Aufit.php .

To install the program just execute the downloaded file.

The program run under Windows Vista SP2, Windows 7 SP1, and Windows 8 or above. The program requires that the Microsoft.NET Framework 4.5 or above is installed. If this is not the case you will be proposed to install it from Microsoft site.

The program is self-updating; it checks for newer versions as they become available. When a new version is available you will be ask if you want to install it. It is usually recommended to accept to install the latest version.

The program uses the ClickOnce technology from Microsoft and therefore it is not installed in a standard location. ClickOnce deployment of Aufit will create a Program Group menu shortcut: "DrCoolZic" -> "Aufit" under Start menu -> All Programs.

Normally the start menu can be found at

```
C:\Users\Your_Login\AppData\Roaming\Microsoft\Windows\Start Menu.
```

The actual program is actually installed inside the following directory.

```
C:\Users\Your_Login\AppData\Local\Apps\2.0\
```

Running the AUFIT Program

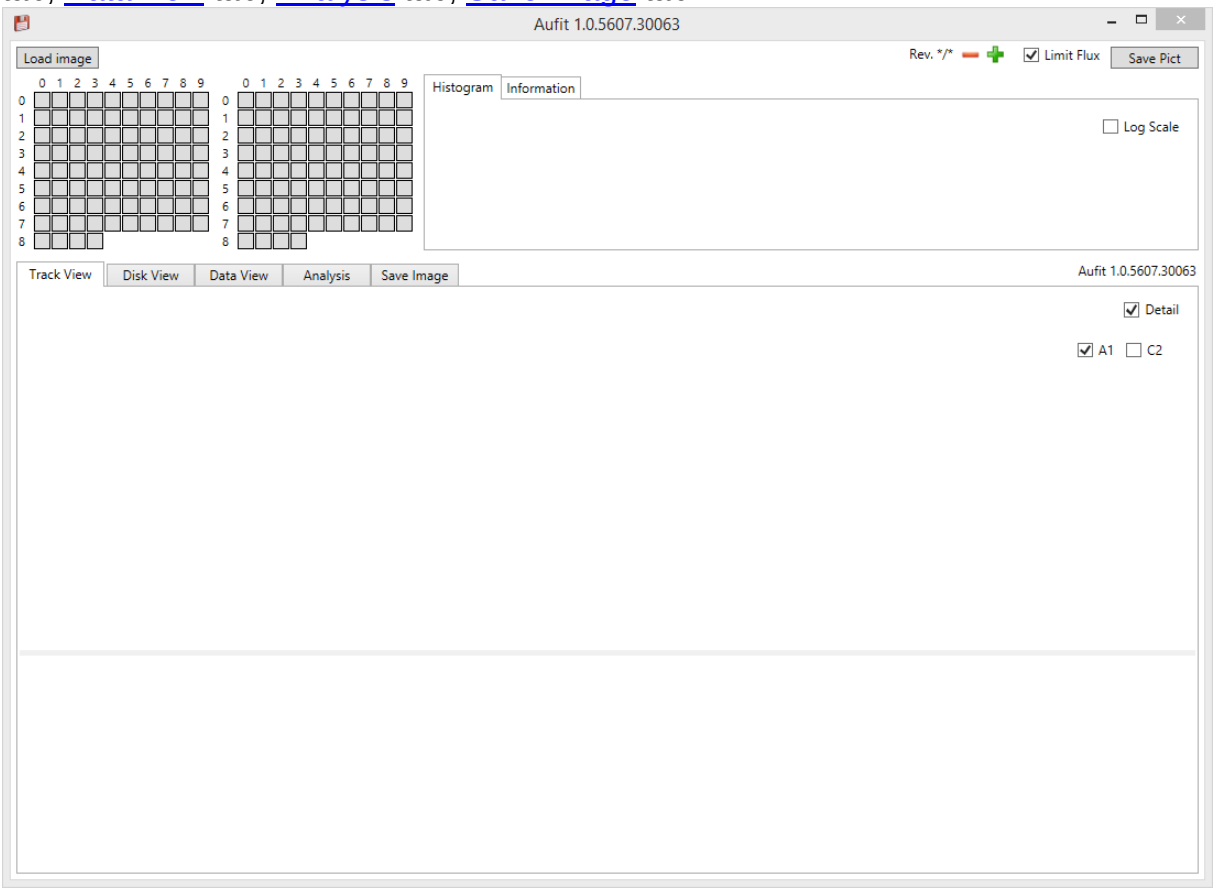
The program can be started by using the Aufit shortcut provided in the start menu. Or by double clicking on an image file associated with the Program.

It is therefore recommended to associate the .**scp** and .**raw** files with the Aufit program (please refer to Microsoft documentation).

When the program starts you will see the following window:

This Aufit windows contains the following three regions:

- On top row region you have:
  - On the left the ***Load Image*** button that allow to specify a FD image to load. Once loaded the name of the file/directory is displayed next to this button.
  - On the right you have the current revolution used ***Rev*** field (as well as the number of revolutions) followed by a **–** and **+** sign used to change the revolution number see Selecting a specific revolution. Followed by the ***Limit Flux*** check box see Flux Transitions Chart for more information on how to use it. Followed by a ***Save Pict*** button. This button allow to save a picture of the currently displayed Aufit window.
- The second row contains the track grid region followed by the Histogram / Information region.
- The third row region contains the different view tabs: Track view tab, Disk view tab, Data view tab, Analysis tab, Save image tab.



Using Aufit

Loading Image Files

Before you can do anything with Aufit you first need to load an Atari floppy disk image. The program reads a complete Floppy disk image that can be composed of a single file (for example a **SuperCopy Pro** .scp file) of a set of files (for example a set of **KryoFlux** .raw files).

In the case of a set of .raw files they must be located in a single directory and they must all have the same "**root**" name (for example awesome) followed by .TrackNum followed by .Side followed .raw. This is the standard naming convention produced by the dtc program. For example the files: *awesome.00.0.raw*, *awesome.00.1.raw*, *awesome.01.0.raw* etc. To read all these files you just need to click any of the raw files in the directory and all the other related files will be automatically read.

There are three ways you can load a floppy disk image in Aufit:

- You can use the ***Load Image*** Button. An open file dialog box will be displayed and you just need to specify the file (or set of files) to load.
- You can drag a floppy image file from the Windows Explorer and drop it into the Aufit program.
- You can double click on an image file associated with the Aufit program (see installation). This will start the program and load the image.

The load operation is perform in two steps:

- First the file is read (IO intensive)
- Then the file is analyzed (CPU intensive)

These two operations are both IO and CPU intensive and therefore can take some times to execute. To maximize performance the program is multi-threaded and will use as many processors / threads available (up to 170 threads).

✎ Note that to fully benefit from the ***Aufit*** stream analysis you need to sample an Atari floppy disk with a minimum of at least **three** revolutions for each track but **five** is the recommended value. Please refer to the imaging tools documentation to create FD images with 3 or more revolutions.

Auto-mode & Command Line Arguments

The Aufit program can be invoked with command arguments, in that case the program run in "auto-mode". The following optional arguments are accepted:

- -disk_view: This argument is used to request Aufit to produce a disk image picture of the FD. When a disk view is requested the optional –scale argument allow to specify a scaling factor for the produce image.
- -scale <scale_value>: This arguments is only useful when the –disk_view argument is used. The scale value is used to scale the image produced. By default the scale is 1.2 but if you specify for example a scale of 2.0 the image produced will have much more details. On the contrary a scale of 0.8 will produce a small image well suited for a Web site.
- -protections: This argument is used to request Aufit to produce a text file that lists all the protections detected for the FD analyzed.
- -save_image: This argument is used to request Aufit to produce a .stx (Pasti) disk image of the FD.
- -suffix <suffix_name>: The "suffix_name" argument is used to compose the name of the output files produced by Aufit in auto-mode (see below).

The arguments must be followed by the name of the input fd image name. If this name contains spaces the image name must be enclosed in double quotes.

Whenever an optional argument is specified on the command line (other than the input file name) the Aufit program is run in Auto-mode that is the operation(s) requested by the arguments are executed and the program terminates after. During execution of Aufit in auto-mode you should not try to execute any command through the GUI of Aufit.

For example the following line:

```
Aufit.exe –disk_view –scale 2.0 –save_image –suffix aufit_1.0 "c:\awesome disk1.scp"
```

Invokes Aufit in auto-mode requesting a disk image picture and a stx pasti file.

The name of the output files produced are the following:

- Disk image: base_name.disk_view.suffix_name.png
- Protections: base_name.protections.suffix_name.txt
- STX file: base_name.image.suffix_name.stx
- An empty file is also produced: base_name.suffix_name.done. This file indicates that the input file has been processed.

The base name depends of the type of the input image.

- For an scp files the base name is directly the scp name. For example if the input is **c:\awesome\awesome_d1.scp** the base name is also **c:\awesome\awesome_d1.scp**.
- Kryoflux raw files are located in a directory like: **c:\awesome\awesome_d1** that contains files track00.0.raw, track00.1.raw, track01.0.raw etc. In that case the base name is c:\awesome\awesome_d1.raw

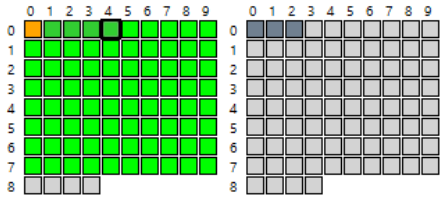
Examples of output:

```
c:\awesome\awesome d1.disk view.aufit 1.0.png
c:\awesome\awesome_d1.scp.disk_view.aufit_1.0.png
```

It is therefore possible to create a dos batch file that finds the floppy disk image to process and invoke Aufit with the right parameters.

As DOS batch files are not easy to write I also provide a program with a nice GUI called **BatchAufit** that automatically invoke Aufit in auto-mode with the appropriate parameters. Please refer to BatchAufit Program for more information.

Selecting a specific track



Once the FD image is loaded the information for the first track of the disk is displayed in the different graphs. In the top left corner of the Aufit window you find two grids (one for each side of the floppy) that are composed of track selection buttons.

Each grid element correspond to a track and the color of the background used gives information about the analyzed track:

- *Lime (green)* for normal tracks,
- *Orange* for protected tracks,
- *Blue* for tracks without sector but that have data following a 3 sync mark sequence. This usually indicates a data track,
- *Yellow* for tracks with possible data but no sync sequence,
- *Slate gray* for unformatted/empty tracks,
- *White* indicates a non-imaged tracks

The currently selected track button is identified with a larger black border around the button. Leaving the mouse over a button display the track number.

You can click on any button of the track grid (except non imaged track) to select a specific track. Once the track grids are "active" (for example by clicking one track selection button) you can change selection from one button to its neighbor by using the four ***direction keys*** (left, right, up, down) of your keyboard.

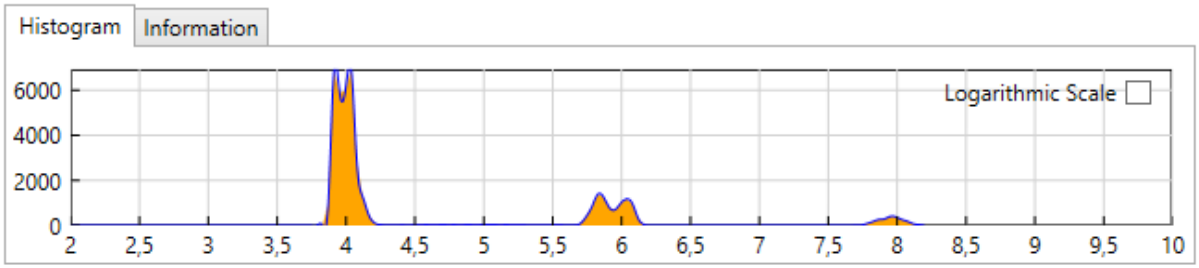
Selecting a new track will update the different views.

Selecting a specific revolution

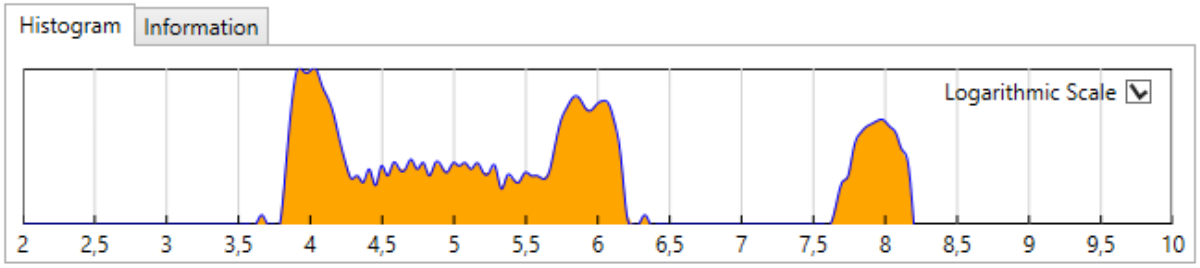
You can select a specific revolution among the one sampled with the **+** and **-** buttons located at the upper right corner of the program window. Selecting a specific revolution affects the information displayed in the different graphs.

The Histogram Graph

The histogram graph, located at the right of the selection grid, represents the repartition of the flux transitions. Normally you should see three pics at 4, 6, and 8 microseconds.

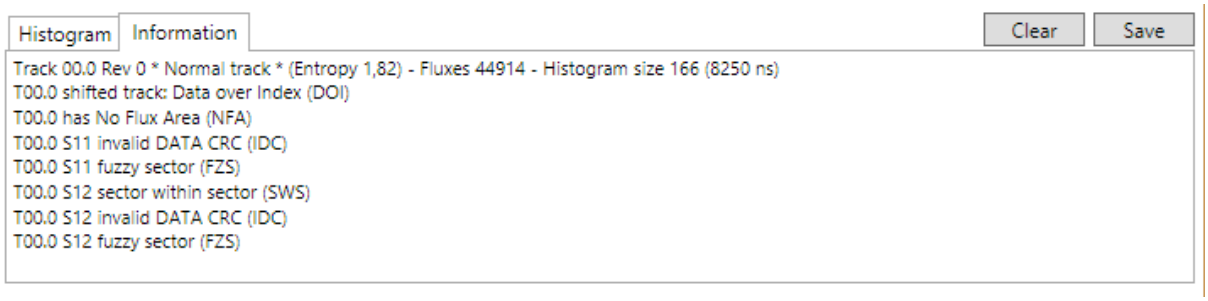


It is sometimes interesting to click the "**Logarithmic Scale**" check box to switch to a logarithmic scale on the Y axis. This allow to see information not visible on a linear scale. For example with the game Dungeon Master you will see that the histogram has a lot of entries between 4 and 6 microsecond due to the "sliding fluxes" pattern used as a protection for this game.



The Information tab

The information tab is used to display information about the read and analyzed track. For example for each track it indicates the entropy, the number of fluxes, and the histogram size (max flux value).

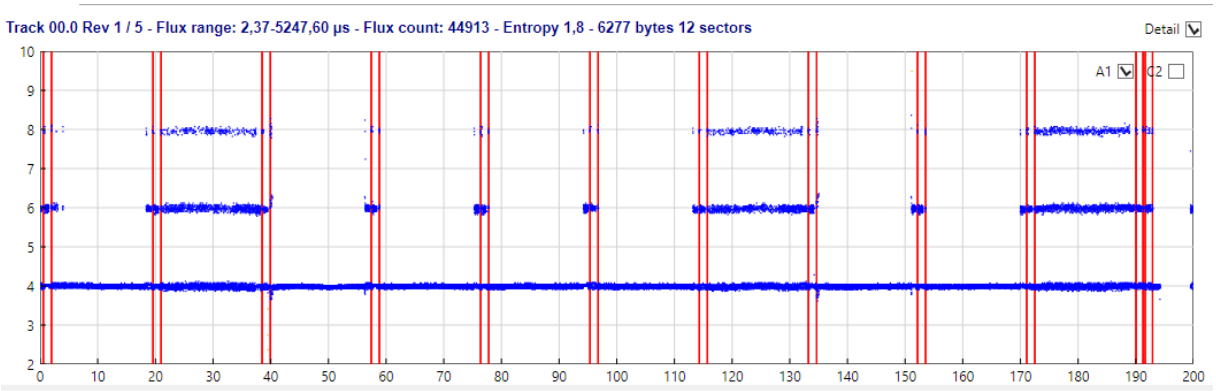


This windows is especially interesting when analyzing the protection. On top of the information mentioned above, each time you select a specific track, the protections used for this track are displayed. The content of this window can be cleared / saved with the corresponding button.

The Track View

Flux Transitions Chart

The unprocessed flux transitions (i.e. flux directly read from the file) are displayed as a ***scattered chart***. A standard track contains about 40000 flux transitions. By default all the flux transitions are displayed. To speed up the display time, for example on a slower machine, you can uncheck the **Detail** check box in upper right corner of the window, this will force to display only 10000 flux transitions (about ¼ of the flux transitions). In the upper right of this chart you have two check boxes called ***A1*** & ***C2***. These check boxes are used to enable/disable the display of the **0xA1** and **0xC2** Sync Marks respectively as they would be detected by the WD1772 using a ***Read Track*** command (i.e. with the sync detector *activated at all time*). You need to refresh the display to see or hide the sync marks. By default the **A1** sync marks (red lines) are displayed and the **C2** sync marks (blue lines) are not displayed.

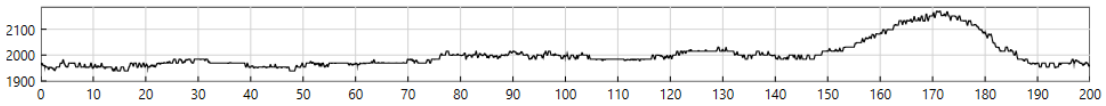


There is a check box button in the upper right corner of the program window called **Limit Flux**. By default this button is checked for unformatted track and is unchecked for formatted track but you can change its state at all time. Both the Histogram chart and the Flux Chart use this checkbox and there content will be updated accordingly.

Limiting flux display is very useful when the track contains long flux transitions (like with NFA) to keep the graph readable. Information about the currently selected track are displayed at the top of the graph.

Bit Width Chart

The Bit Width chart displays the width of decoded half bit as computed by the internal DPLL clock. The nominal bit width is 2000ns but of course this value can vary a lot for over / under clocked sectors / tracks.



This bit width is computed by the DPLL and is proportional to the running average of the flux transitions fed to the DPLL. The Y scale is automatically adjusted.

The Bit Width normally stays relatively stable around the 2000ns period but for protection like Copylock the values usually goes from 2000 to 2100 (5% increase) inside a sector. Some protections over clock or under clock the complete track.

In some specific cases like unformatted track/sector the period can go up to 2200ns or down to 1800ns, reaching the 10% limits hardcoded in the DPLL.

Data Graph

This graph displays all the sectors found on a track:

- The ID field is displayed in yellow (you probably need to zoom to see it),
- The Intra-gap field is displayed in light green,
- The Data field is displayed in blue.



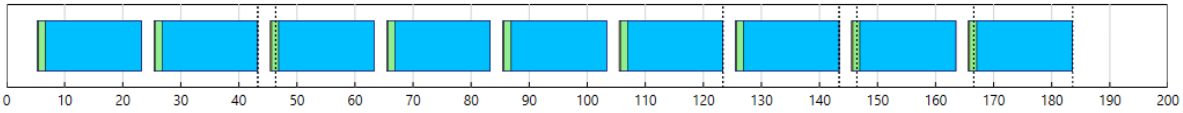
The data field color can change from Blue to Aqua (lighter blue) for under clocked sector, to Deep Sky Blue for overclocked sectors, and to Purple for sector with bit width variation (e.g. speedlock).

Data sectors with fuzzy bits have an orange border and data sectors with CRC error have a red border.

If you place the cursor over a sector in the data graph, detailed information about the sector is displayed.

The Data graph information uses a complex algorithm (mixture of read track, read address, and read sector) to display correctly "complex" tracks containing protection like: Sector within sector, tracks with large number of sectors, tracks with one sector, etc.

It is not easy to see "out of band" transitions in the flux transition graph. Therefore "out of band" transition positions are displayed as a dotted line in the data graph. When this happen in the inter-gap and at the end of the data gap this may be sector-splice indicating that the sector have been written on an Atari.



Zooming and Panning in the Track view

When the **flux chart** is not zoomed (default when first displayed) you can use the left mouse button to define a zoom area (zoom in both in x and y directions). Click on a first point to start the definition of the area, keep the mouse button down and move to a second point to define a zoom area. A rubber band box is drawn that delimitates the area and when the left button is released the graph is zoomed in the area defined. The scales displayed on both axis are automatically adjusted function of the zoom factor.

Once zoomed a click with the left button allow to pan the graph. Click any point in the flux graph with the left button and while keeping the button down move the cursor. The image of the graph dynamically moves with the cursor. When you release the mouse button, the graph and the axis position are updated.

You can further **zoom-in** or **zoom-out** by using the mouse wheel.

If you want to reset the graphs to un-zoomed state click anywhere in the Flux chart with the right mouse button (this is also the way to force a redisplay).

Note that the same zoom factor is used to update the **flux chart**, the **clock chart**, and the **data chart**. Therefore these three graphs always stay synchronized.

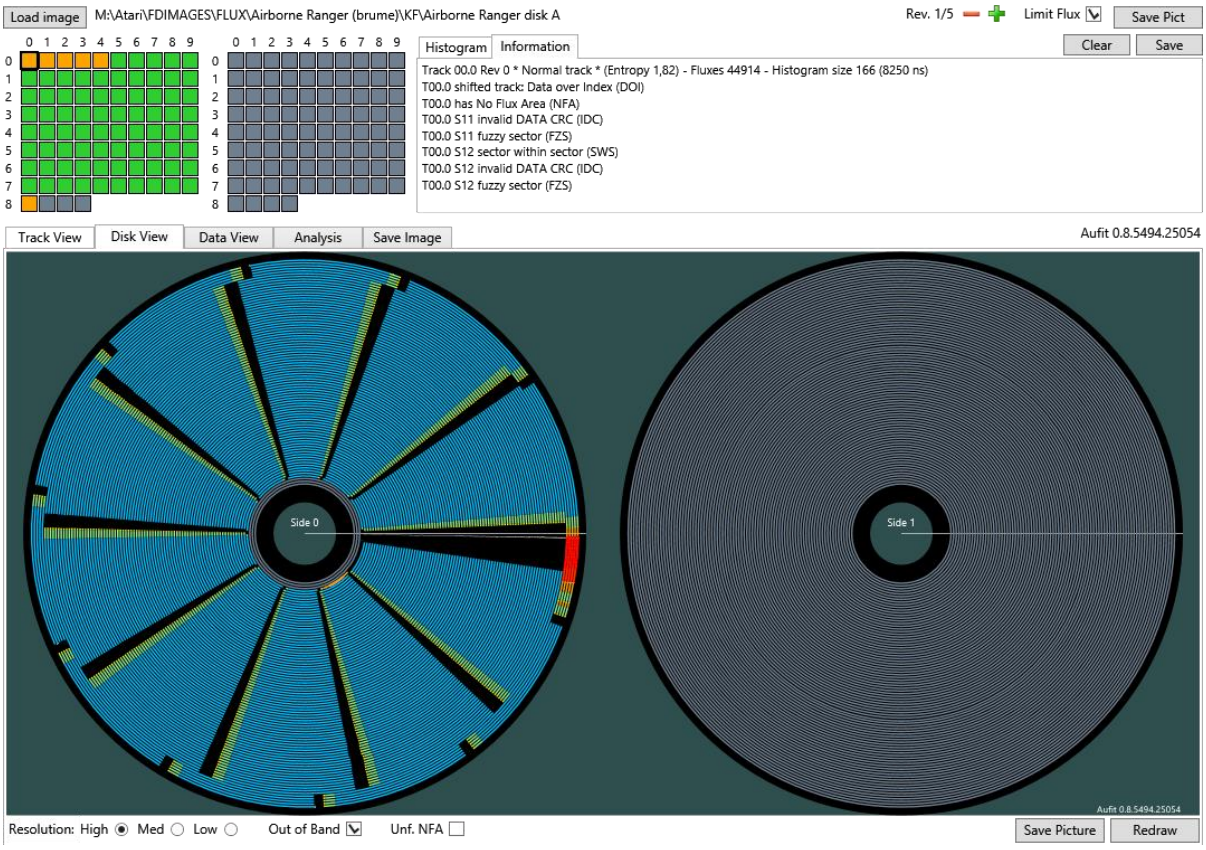
You must place the mouse on top of the **flux chart** to pan and zoom the charts.

The Disk View

The **Disk View** tab allow to display the layout of the two sides of the floppy disk.

The following color information is used:

- The ID field of a sector is in Yellow (hard to see)
- The Intra-gap of a sector is in light green
- The Data field of a sector is displayed:
  - In Deep sky blue by default
  - In Blue for over clocked sector
  - In Aqua for under clocked segment
  - In Purple for sector with bit-rate variation (e.g. speedlock)
- Data sector with CRC error or Fuzzy bytes are displayed in Orange
- No Flux region are displayed in Red
- Unformatted track in Gray
- Data track in Blue
- Track with "data in band" but no sync in Yellow
- Data bits outside normal range are displayed in White



It is possible to save the layout picture with 3 resolutions (high, medium, low).

Selecting a track highlight the track in the Disk View.

The Data View

The data view contains a ***MFM*** check box on bottom left. This check box allow to display an extra line with MFM word. The ***Save*** button allow to save the content of data view in .rtf format. This format allow to keep the formatting information (color, font etc.) and it must be loaded with a program like Microsoft Word. The ***Layout, Track, Sectors*** buttons are used to display the information as explain below.

Track Layout

The track layout allow to see the layout of the currently selected track in an array. This array is displayed if you click the ***Layout*** button.

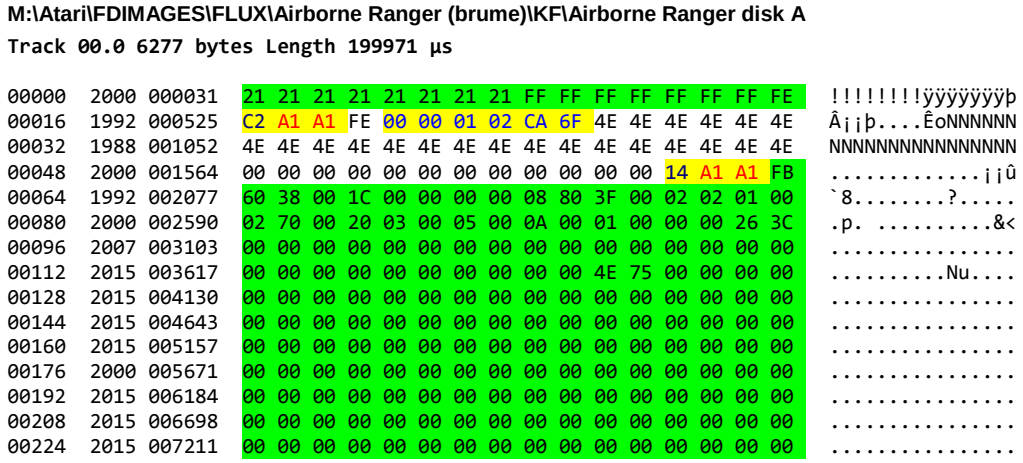
M:\Atari\FDIMAGES\FLUX\Airborne Ranger (brume)\KF\Airborne Ranger disk A
Track 00.0 6277 bytes has 12 sectors

ID Field			GAP	Data Field					GAP
Sector	Position	CRC	Bytes	Size	Position	Time	CRC	Fuzzy	Bytes
1	637	OK	37	512	2043	16482	OK	No	35
2	19641	OK	37	512	21038	16415	OK	No	35
3	38566	OK	38	512	39989	16401	OK	No	34
4	57498	OK	37	512	58900	16445	OK	No	35
5	76460	OK	37	512	77857	16443	OK	No	35
6	95421	OK	37	512	96827	16438	OK	No	35
7	114378	OK	37	512	115775	16405	OK	No	35
8	133296	OK	38	512	134724	16415	OK	No	34
9	152222	OK	37	512	153621	16426	OK	No	35
10	171161	OK	37	512	172560	16424	OK	No	35
11	190103	OK	31	512	191315	16480	BAD	Yes	-495
12	191700	OK	33	512	192972	16487	BAD	Yes	

- We first have information about the ID field:
  - The sector number
  - The start position of the ID field in microseconds
  - A flag to indicate if the CRC of the ID field is good or not
- Then we have the number of bytes in the GAP located between the ID and Data field. This number includes the post-id gap, the pre-data gap including the 3 sync marks. For more information refer to Standard 9-10-11 Sectors of 512 Bytes Format
- Then we have information about Data fields
  - Number of bytes of the sector
  - The start position of the Data field in microseconds
  - The length of the sector in microseconds.
  - A flag to indicate if the CRC of the Data field is good or not
  - A flag to indicate if the Data field has fuzzy bytes
- Then we have the number of bytes in the GAP located between a sector and the next sector. This number includes the post-data gap, the pre-id gap including the 3 sync marks. For more information refer to Standard 9-10-11 Sectors of 512 Bytes Format. A negative value indicates a sector within sector.

Track Buffer

The track buffer displays the content of the selected track as they would have been read by the ***Read Track Command*** of the WD1772. The buffer is displayed if you click the ***Track*** button.



The data bytes are displayed by group of 16 with the following information:

- The first number is the position of the byte in the buffer (decimal value)
- The second number is the half bit-width of the first byte of the line. If the width is 2.5% above normal the period is displayed in blue, if the period is 2.5% below normal it is displayed in red.
- The third number is the starting position of the first byte of the line in microsecond relative to the index of the track.
- 16 bytes are displayed in hexadecimal format
- The corresponding 16 ASCII characters for the 16 bytes.

Colors and font style are used to provide additional information.

Background color:

- Light blue for invalid characters (F5-F7) into gaps
- Coral for no flux area (MFM 0x0000 / 0xFFFF)
- Yellow for Sync characters (MFM 0x4489 and 0x5224)
- Light green for bytes located inside an ID field or a Data field.

Character color:

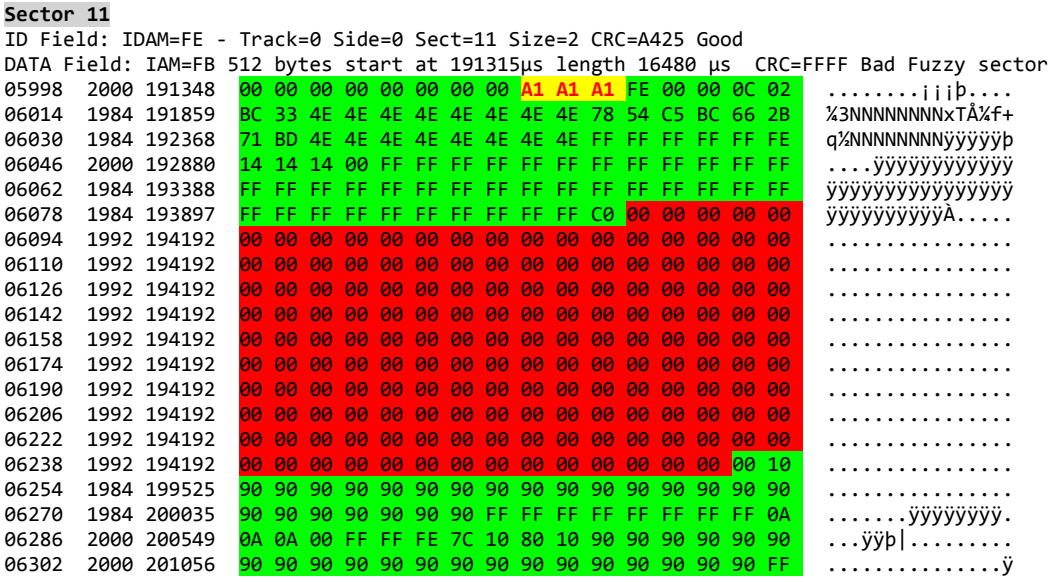
- Red for 0x4489 Sync characters (0xA1)
- Navy blue for 0x5224Sync characters (0xC2)
- Blue for ID bytes
- Black for Data and intra-gap byte and for inter-gap byte

Bold characters:

- For Sync characters (MFM 0x4489 and 0x5224)
- For Id field content

Sectors Buffer

The sector buffer displays the content of the selected track as they would have been read by the ***Read Address*** and ***Read Sector*** commands of the WD1772. The sector buffer is displayed if you click the ***Sector*** button



For each sector we have

- A line indicates the IDAM value, the ID content (Track, Side, Sect Number, Size, and CRC) and if the CRC is good or not
- A line indicates the IAM value, the number of bytes of the sector, the position in microseconds, the length in microseconds, the CRC value, a flag to indicate if the CRC is good or not, and a flag to indicate if the sector contains fuzzy bytes.

This is followed by a dump of all the data bytes exactly as they would have been read by the ***read sector*** command of the WD1772. The data bytes are displayed by group of 16 with the following information:

- The first number is the position of the byte in the track (decimal value)
- The second number is the half bit cell width of the first byte of the line. If the period is 2.5% above normal the period is displayed in blue, if the period is 2.5% below normal it is displayed in red.
- The third number is the position of the first byte of the line in microsecond relative to the index of the track.
- 16 bytes are displayed in hexadecimal
- The corresponding 16 ASCII characters for the 16 bytes.

Color is used as explained in Track Buffer.

For example here we can see that the sector contains 3 sync marks displayed with a yellow background (sector within sector) and further an NFA region displayed in red.

If fuzzy bytes have been found for a sector a "fuzzy mask" buffer is also displayed.

To find fuzzy bytes the Aufit program read the same sector multiple times from the different revolutions. If the content of a byte in the same position is different the mask value is the result of a **xor** operation.

```
Mask[offset] = (Sector_read_1[offset] ^ Sector_read_2[offset])
```

Thus same value (1&1 or 0&0) gives a result of zero and different value (0&1 or 1&0) gives a result of one. For example if the same byte is read one time as 0000 1111 and the next time as 0101 0101 the mask will be 0101 1010.

The mask can then be used to emulate fuzzy bits as follow:

```
FuzzyByte[offset] = (Sector[offset] & Mask[offset]) | (rand() & Mask[offset])
```

Where rand() is a function that randomly returns true or false.

The information for a sector presented above repeats for all sectors.

Saving Data View content

For any of the data view buffer (Layout / Track / Sectors) it is possible to save the buffer in RTF format using the ***Save*** button.

The RTF format allow to keep all formatting information for example the font weight, the color, etc.

You can easily use a reader like Microsoft Word to load the RTF format and save it for example in PDF.

The Analysis View

Currently the Analysis view only contains the protection analysis. Other analysis will be added at later time.

To perform a protection analysis just click the ***Protections*** button. The result of the analysis of all the track is displayed:

To save the result of analysis in text format you can use the ***Save*** button.

```
Protection analysis for M:\Atari\FDIMAGES\FLUX\Dragonflight\SCP\Dragonflight - A.scp

T00.0 hidden data in GAP (dragonflight)
T00.0 S1 invalid head/side number 48 (IHN)
T00.0 S2 invalid head/side number 48 (IHN)
T00.0 S3 invalid head/side number 48 (IHN)
T00.0 S4 invalid head/side number 48 (IHN)
T00.0 S5 invalid head/side number 48 (IHN)
T00.0 S6 invalid head/side number 48 (IHN)
T00.0 S7 invalid head/side number 48 (IHN)
T00.0 S8 invalid head/side number 48 (IHN)
T00.0 S9 invalid head/side number 48 (IHN)
T00.0 S10 invalid head/side number 48 (IHN)
T01.0 hidden data in GAP (dragonflight)
T01.0 S1 invalid track number 178 (ITN)
T01.0 S1 invalid head/side number 48 (IHN)
T01.0 S2 invalid track number 178 (ITN)
T01.0 S2 invalid head/side number 48 (IHN)
T01.0 S3 invalid track number 178 (ITN)
T01.0 S3 invalid head/side number 48 (IHN)
T01.0 S4 invalid track number 178 (ITN)
T01.0 S4 invalid head/side number 48 (IHN)
T01.0 S5 invalid track number 178 (ITN)
T01.0 S5 invalid head/side number 48 (IHN)
T01.0 S6 invalid track number 178 (ITN)
```

The following protections are currently analyzed.

- Track Protections
  - Short track
  - Long track
  - Bad data into GAP
  - Data beyond index
  - Data above index,
  - ID beyond index,
  - ID above index
  - NFA
  - Non sequential sector number (for ST format)
  - Nonstandard sector size (for ST format)
  - "Dragonflight & Jupiter Masterdrive protections (hidden data into gap)
- Sector Protections
  - Duplicate sector number
  - Nonstandard IDAM
  - Invalid track number
  - Invalid Side Number
  - Invalid Sector number
  - Invalid sector size
  - ID CRC error
  - Sector within sector
  - Data CRC Error
  - Fuzzy Sector

- Nonstandard DAM
- Long sector
- Short sector
- Sector bit width variation
- Sector with no data

For more details about these protections please refer to my document **Atari Floppy Disk Copy Protection**.

✎ Note that protections are also analyzed in other views when a Track is selected. You can see the result in the Information window. Remember that a track that contains protection or error is displayed in Orange in the track grid.

Save Image View

When you select this tab you are presented with information about the possible format you can use to save the current image.

Currently two families of formats exist:

- The non-protected formats includes: ST, MSA, DIM
- The protected format: STX

Not all formats are yet supported.

✎ This version of Aufit includes an auto-recovery feature. If you try to write the content of a floppy disk that contains sector's errors the program will try to recover the information about this sector from another revolution.

You first have to select the output format you want to use from a combo box on the left. If it is not possible to use a specific format with the current data (for example ST format for floppy disk using protection) it will be grayed.

When you click the ***Write Image*** button a form open and ask you the location and the name of the file to write. If the file already exist a window will pop up and ask you if you want to replace the existing file.

As soon as you close this form the output file is written and some information is displayed in the main window including error recovered and potential problems.

✎ Reminder: In order to produce a correct file you must use images with data for at least three revolutions but a value of five is recommended especially for floppies in bad condition.

If not enough data (revolutions) have been imaged a window will pop up and warn you. You are still offer the capability to force writing the file.

Normally single sided / double sided detection is performed automatically by Aufit. If you are getting error messages in Steem that an image of a track is missing on side 1 of the floppy image you may need to force Aufit to write file for double sided disk. For that matter please check the ***Force DS*** check Box.

When the ***Optimize*** check box is checked (the default) Aufit will write "compressed" file (smaller file) for the STX and MSA format.

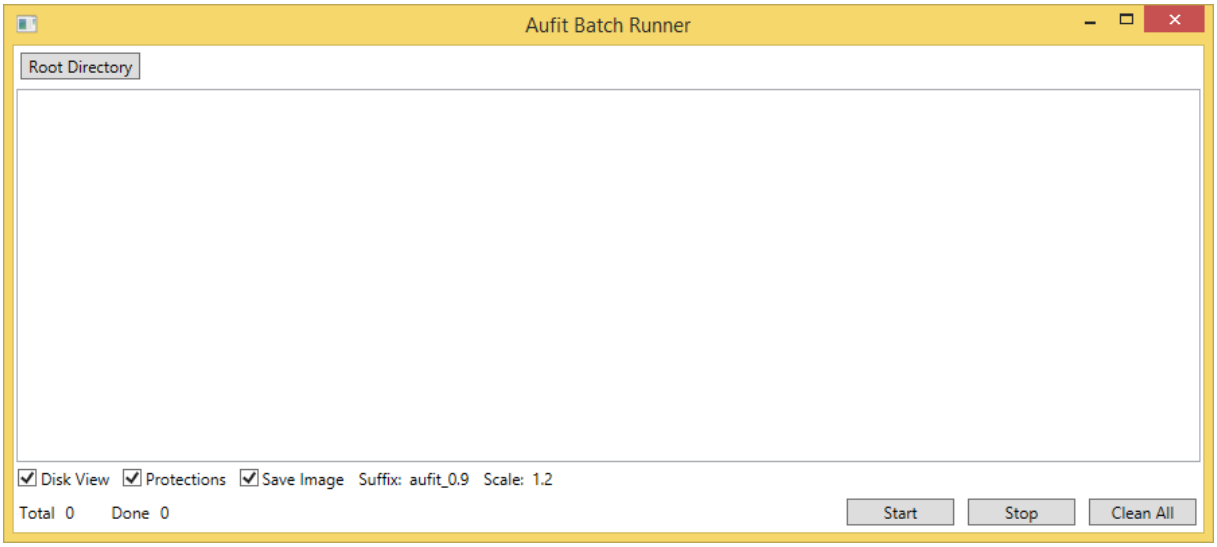
The ***Debug*** and ***WriteSync*** check boxes are for debug purpose and normally you do not have to use them.

BatchAufit Program

As mentioned in the Auto-mode & Command Line Arguments section it is possible to run the Aufit program in batch mode to produce specific outputs.

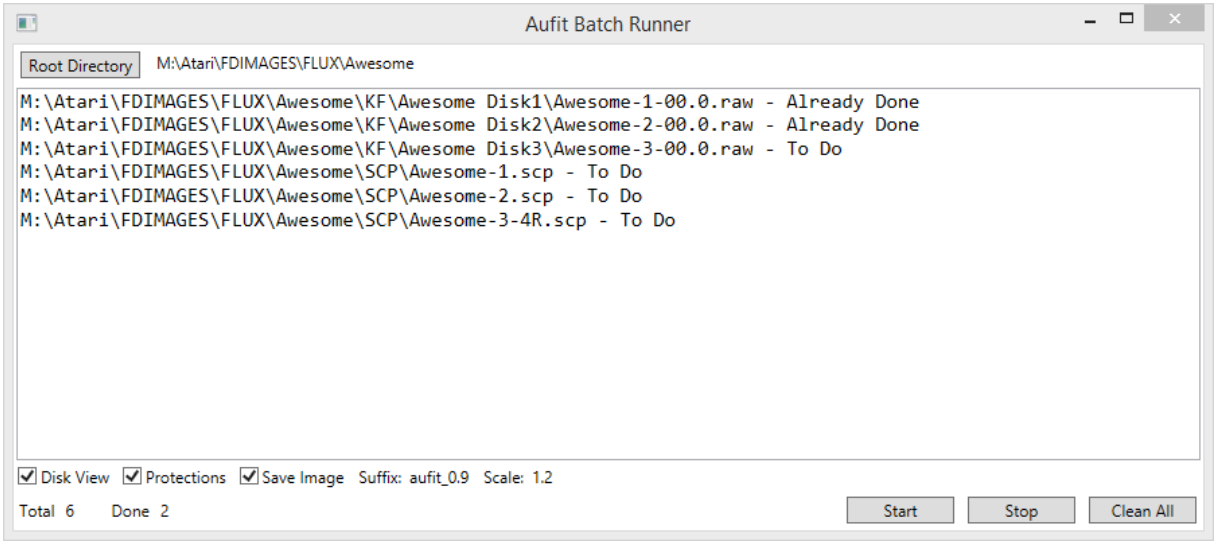
The BatchAufit program allow to automatically detect all FD images located inside a directory tree and run Aufit in auto-mode for all these files.

The GUI is composed of three area



The top area contains a button that allow to specify the root directory of the tree where you want to search for the Kryoflux and SCP FD images. When the "root directory" button is clicked a dialog box opens that allow to specify the root directory.

As soon as this root directory is specified the tree is analyzed and all the Kryoflux and SCP images are detected and displayed in the middle area.



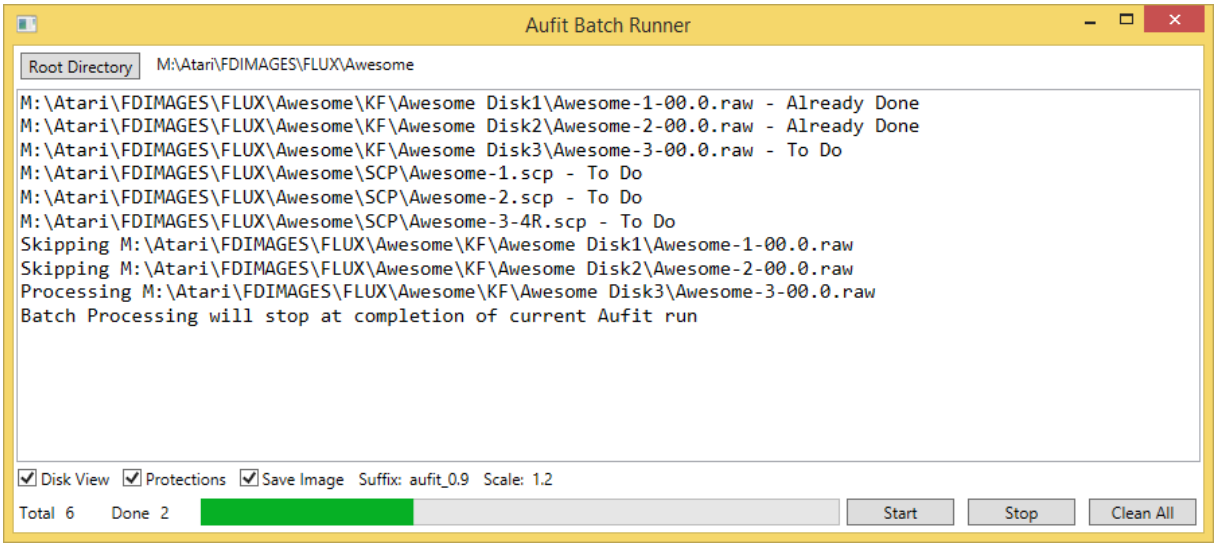
An indication of the Total number of FD images as well as an indication on the number of already analyzed images is displayed. In the above example we can see that 6 images have been detected (3 KF + 3 SCP) and that 2 of them have already been processed (an already processed file is indicated by a .done file).

Just below the middle area you will find several checkboxes that allow to specify if you want to create disk_view, protections, and stx images when running the Aufit program in batch mode. This is followed by to text boxes that allow to specify the

suffix to be used when creating the output files, and the scale to be used when creating disk view files.

At the bottom of the GUI are three buttons:

- The **Start** button allows to start the batch execution of Aufit. A progress bar indicates the percentage of images already processed.
- The **Stop** button allow to stop the execution of the batch process. The execution of Aufit on a current image is terminated and the batch process stops. As the program can detect what has already been done it is possible to restart the execution where it was stopped (even in different run of BatchAufit).
- The **Clean All** button removes all the output files created by running Aufit in batch mode. You should use this command with care!



Sometimes you may want to ignore specific subtrees inside the directory tree that you want to analyze. In that case you may place a file called **aufit.ignore.txt** to indicate that you do not want the current directory and all its subdirectories to be analyzed.

Checking Quality of generated images

This section discuss several questions around the quality of Floppy Disk images created with devices like KryoFlux or SuperCard Pro.

To some extent the quality of the image is only related to the floppy drives and floppy disks conditions and therefore independent of the imaging device used.

Cleaning

The first thing you want to do before imaging floppy disks is to clean the floppy drive head and a make sure you have spotlessly clean floppy disk.

Cleaning the floppy drive head and cleaning the diskette can make a big difference

First always clean the floppy drive head using dedicated cleaning kits as this is a very easy to do operation.

Cleaning a floppy disk is a much more painful operation and therefore you can try to image the diskette without cleaning. If during analysis, as explained below, you find problems on the disk image you should clean the floppy diskette. The recommended way is to wipe the surface of the floppy diskette using a Q-Tip impregnated with head cleaning liquid. Normally this liquid evaporate quickly but you can use the other side of the Q-Tip to help dry the liquid in excess. There are numerous site and videos on the net that explain in detail these operations.

Important things to know when cleaning a floppy diskette:

- The side with an etiquette is the "second logical side" (side/head 1) of the diskette
- The side with the hub that engages with the drive motor is the "first logical side" (side/head 0) of the diskette.

So if you find problems on side/head 0 you need to clean the side without the etiquette (what I refer as the back side) and if you find problems with side/head 1 you need to clean the side with the etiquette (what I refer as the front side).

Analyzing Images

Now let's move on the heart of the subject:

- How do I find if the image I have done is of good enough quality?

For that matter we use the ***Aufit*** program.

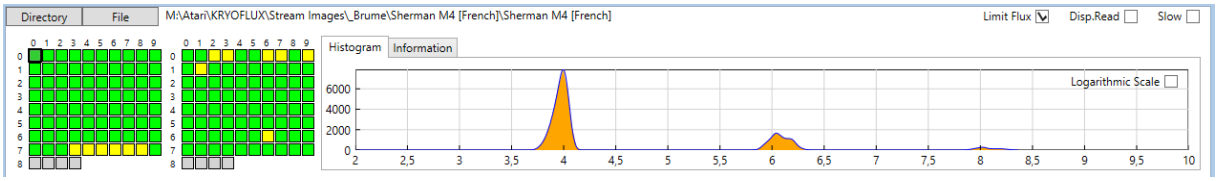
You first need to read the image you have created as explain in Reading Image Files section.

When the program read the image it already display useful information in the Track grids as the background color of each track is an indication of possible problem:

- A "normal" track has a light green background,
- an unformatted track has a light gray, and
- a track with potential problem is displayed in yellow.

At this stage only a quick Shannon Entropy analysis is done on the flux transitions histogram and a yellow color usually indicates track with partially unformatted areas or incorrectly imaged track but always require a more complete analysis.

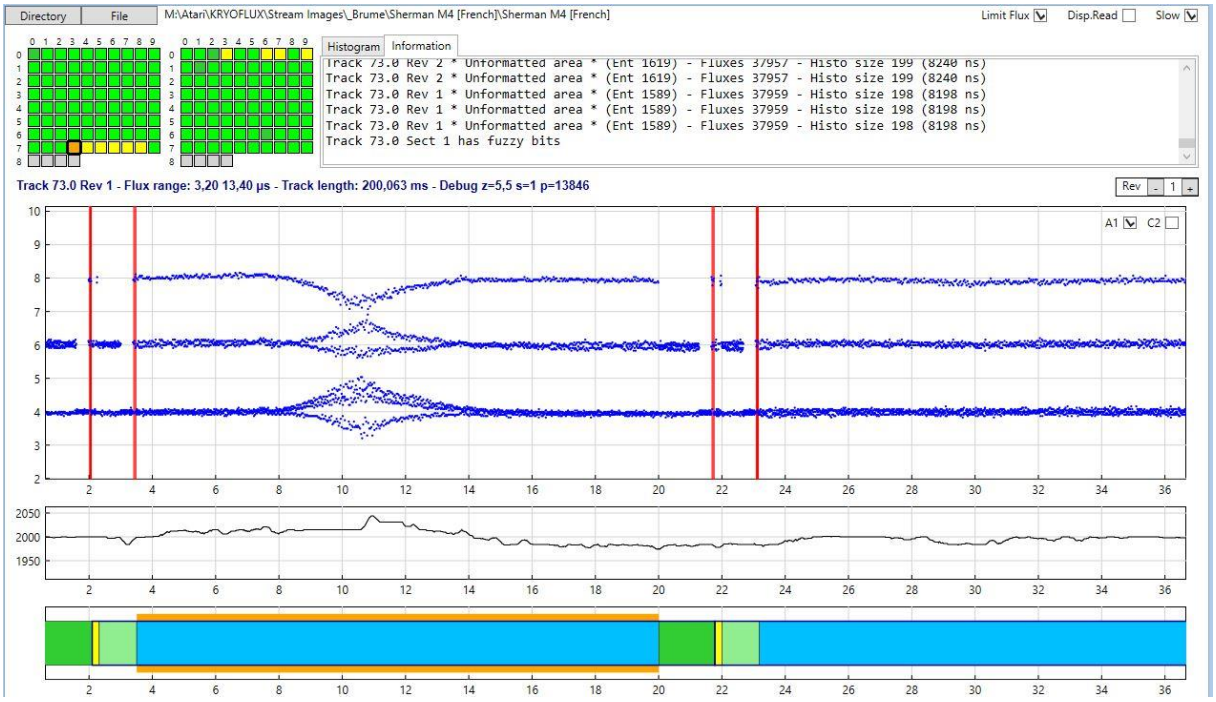
For examples here is the result of reading an image of Sherman M4.



We can see that we have several tracks with a yellow background. If we click on some of these tracks we will find that in fact they are corrects.

But if we click on track 66 we can see than sector one has a lot of "dirty" flux transitions. Thanks to the DPLL these variations can be corrected and we are able to read the sector correctly. This is indicated by the fact that all the sectors read without CRC error.

But if we select the track 73 and zoom on the first sector area we can see that we also have dirty flux transitions, but here the DPLL is not capable to decode the information correctly.



In this case we can see that the first sector is read with fuzzy bytes (orange box in data chart and indication in the information window). In fact if we further analyze the problem we can find, in the fuzzy mask buffer that only **two** bytes at position 206 and 214 vary between revolutions! This also why it is recommended to sample at least four revolutions as **Aufit** always checks fuzzy bytes by comparing the data read for all available revolutions.

```
Fuzzy bytes for sector 1
00000  2000 003505  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00016  2000 004013  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00032  2015 004526  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00048  2015 005040  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00064  2005 005554  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00080  2015 006068  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00096  2010 006582  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00112  2018 007097  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00128  2031 007614  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00144  2005 008128  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00160  2005 008644  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00176  2015 009160  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00192  2015 009676  00 00 00 00 00 00 00 00 00 00 00 00 00 00 01 00  ................
00208  2015 010192  00 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00  ................
00224  2020 010709  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

In this case it seems obvious that is error is not part of a protection but due to a problem on the floppy diskette. In such case you should try to clean the diskette and image it again.

We can even look further at the problem. If we click the **Sectors Buffer** command we can see that on some of the revolutions the CRC is good and on some other revolutions the CRC is bad. Normally a fuzzy byte always result in a CRC error. Therefore this indicates that on some revolutions we have read the correct information. Currently Aufit is not able to use this information to create a working Past file but I may implement this feature in the future.

- You need to be aware that on an Atari system when reading sectors from a diskette an automatic retry is performed anytime a sector is not read correctly. This retry is transparent to the user and is used more often than you think.
- Therefore when you have problem clean the floppy and recreate new image, and if this still fails try to recreate the image several times. If the floppy is only marginally not read this may allow you to create a good image.

In order to find all possible problems you first need to run the **Protection** command and for each "protected track" (displayed with a red background in the track grid) you need to check if the problem comes from known protection or from "dirty flux transitions". Unfortunately it seems very difficult to find out about bad images automatically and this is why you need to do a visual inspection using the different features of the Aufit program.

Document History

- V1.0 – May, 2015
- V0.9 – January, 2015 – Preliminary documentation for Aufit 0.9
- V0.4 – March 1, 2014 – Document double sided disk layout, Force DS, and color scheme changes.
- V0.3 – February 17, 2014 – Updated information for release 0.3a
- V0.2 – January 18, 2014 – Added information about usage of the program, about protection analysis, and Writing Pasti files. Added a new complete section about analysis of the quality of the imaged floppy disks.
- V0.1 - January 9, 2014 – Preliminary release for v0.1 Alpha version.
- V0.0 – January 6, 2014 - preliminary first release.

Program History

- V1.0 May 2015
  - Aufit batch mode
  - Release of BatchAufit
  - Display of Out of band flux in Data chart
- V0.9 February 2015
  - Support for MSA and ST output format
  - Detection of Drzgonflight & Masterdrive protections
  - Improved display of several charts
- V0.8 January 2015 - Data View
  - - Bug fix: DSR + FDC (see todo)
  - - GUI cleaning + Bug fixes (GAP - FDC)
  - - removed file association create too many problems
  - - Save Window size and position
  - - Test Click once publishing
  - - Buffer Windows are gone replaced by the Data View tab with better functionalities.

- V0.7 January 2015 - Auto recovery - Improve Disk view - Lots of bug fix - Auto safe mode for really bad images
  - - Added auto safe mode (safe but slower mode) when sector not found + Analysis GUI + Speed DiskGui + Bug fixes
  - - Fix DSR bug on sync seq, FDC detect sync at edge of index + bug fixes
  - - FIXED 0.6.5476.33179: Crackdown 2D = KF Sector not found
  - - FIXED 0.6.5476.18307: Computer Hits V2 - SCP D2 sector missing on some tracks
  - Crackdown SCP D1&D2 crash
  - - Auto-recovery feature added to automatically write good pasti
  - - FIXED 0.6.5473.34262: Chicago 90 crash
  - - FIXED 0.6.5472.17463: Gunship SCP track 21.0 Bad Fuzzy report first sect due to NFA!
  - - FIXED 0.6.5473.16121: Barbarian scp d2 invalid file content error cannot display
  - - Fix bug Kickoff 2 SCP NFA not decoded correctly (again cant display right info)

- - Fix bug NFA not working for Dr. Doom Revenge (now NFA handled at read time)
- - *** NOTE *** SCP file does not provide information about index position in flux
- Therefore it is impossible to do something clean when we have NFA over index
- - Fix bug sector not found due to unformatted sectors Crossbow D1, RoadBlaster
- - improve Disk View (cursor display) & speedup Track View

- V0.6 December 2014 Freeze for Beta release to Brume
  - - Beta release

- V0.5 December 2014 Major rewrite of Aufit internals
  - - Fix non working STX for Jupiter master drive, Gunship, P-47
  - http://www.atari-forum.com/viewtopic.php?f=102&t=25854&start=525#p263646
  - http://www.atari-forum.com/viewtopic.php?f=102&t=25854&start=250#p248282
  - http://www.atari-forum.com/viewtopic.php?f=102&t=25854&start=275#p248377
  - - Fixed nasty bug in DSR for Skateball game
  - - New Disk Layout display design
  - - New Main window layout using tabs for: Track view, Disk view, Content, Analysis, Save
  - - Specify file/dir to read = from menu, drag and drop, double click
  - - Drawing disk layout now uses multi-threading
  - - now reading and processing of flux data is using heavy multi-threading
  - - Added out of band flux display in Disk layout window
  - - code and documentation cleaning
  - - detection of ID over index on top of data over/beyond index
  - - optimization of STX file using data from track for sector (save space)
  - - support for stream file drop in main window
  - - protections rewritten completely - uses new flags on new DS
  - - capability to save pictures of main screen and Disk layout
  - - Handle correctly intra sector bit width variation
  - - FDC speed improvement: all sectors read in one pass in most cases
  - - removed read directory and dependency on Ookii.Dialogs.Wpf.dll
  - - This version uses new "packed" data structures to be prepared to interface with IPF lib.
  - - now flux samples are stored in ns and speed correction is done at reading time
  - - Completely rewritten FDC module with more clean routines
  - - use modified version of the DSR from IPF library thanks to IFW from SPS
  - - changed clock reference to half bit width
  - - removed rev=0 capability

- V0.4b - March 1, 2014 Color scheme changed, improve disk data, fix pre sync data problem, and fix fuzzy assertion error in compare mode.
- V0.3a February 17, 2014 Check for incomplete information when writing Pasti. Corrected NFA display, improved FDC emulation. More robust: should be able to read corrupted Floppy Disk
- V0.2c January 22, 2014 – Bug fixes on Pasti Files generated
- V0.2a January 18, 2014 – Major change is the capability to write Pasti files. This is still experimental. Bug fixes and better read track report. Now Buffer display uses the currently displayed revolution. This allow to compare the decoded data for the different revolutions.
- V0.1b January 9, 2014 – Fixed one very nasty and tough to find bug. In .Net framework when you click a form to select a file the click event is not flushed and you get it back as a button click ... Also improved memory usage.
- V0.1a January 9, 2014 – Compared to initial release Protection analysis has been added. So only missing initial features is Pasti write capability.
  Known problems: Run out of memory easily, GUI freeze with DISK DATA command.
- V0.0 January 6, 2014 – Initial alpha release on request.